

Assignment: NumPy, Pandas & Matplotlib

Difficulty Level: Beginner ★

Instructions:

- Complete all questions using Python.
 - Write clean and well-commented code.
 - Use Google Colab or Jupyter Notebook.
 - Submit a single `.ipynb` notebook containing all solutions.
 - Display outputs for every question.
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Section A – NumPy (8 Questions)

Q1. Create a NumPy array

Create a NumPy array containing numbers from **1 to 20** and display it.

Q2. Array Statistics

Using the array created above, find:

- Maximum value
 - Minimum value
 - Mean
 - Median
 - Standard Deviation
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Q3. Array Reshaping

Create an array of numbers from **1 to 16** and reshape it into a **4 × 4** matrix.

Q4. Arithmetic Operations

Create two NumPy arrays of equal size and perform:

- Addition
 - Subtraction
 - Multiplication
 - Division
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Q5. Even Numbers

Create an array from **1 to 50** and display only the even numbers.

Q6. Random Numbers

Generate an array of **15 random integers** between **10 and 100**. Find:

- Maximum
 - Minimum
 - Average
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Q7. Array Indexing & Slicing

Create an array from **1 to 30** and display:

- First 10 elements
 - Last 10 elements
 - Elements from index 5 to 15
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Q8. Matrix Operations

Create two **3 × 3** matrices and perform:

- Matrix Addition
 - Matrix Subtraction
 - Element-wise Multiplication
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Section B – Pandas (8 Questions)

Q1. Create a DataFrame

Create a DataFrame containing the following columns:

- Student Name
- Age
- Course
- Marks

Add data for at least **5 students**.

Q2. Dataset Exploration

Using the DataFrame:

- Display the first five rows.
 - Display the last five rows.
 - Display the column names.
 - Display the shape of the DataFrame.
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Q3. Basic Statistics

Find:

- Average Marks
 - Highest Marks
 - Lowest Marks
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Q4. Filtering Data

Display students who scored **more than 80 marks**.

Q5. Sorting Data

Sort the DataFrame in **descending order of Marks**.

Q6. Add a New Column

Create a new column named **Result**:

- Pass if Marks ≥ 40
 - Fail otherwise
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Q7. GroupBy Operation

Create a DataFrame containing at least **10 students** from **3 different courses**.

Find the **average marks** for each course.

Q8. Sales Data Analysis

Create a simple sales DataFrame with:

- Product
- Category
- Quantity Sold
- Sales Amount

Find:

- Total Sales
 - Best Selling Product
 - Average Sales Amount
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Section C – Matplotlib (8 Questions)

Q1. Line Chart

Create a line chart showing monthly sales for **six months**.

Q2. Bar Chart

Create a bar chart showing sales of different product categories.

Q3. Pie Chart

Create a pie chart showing the percentage of students in different courses.

Q4. Histogram

Generate random student marks and create a histogram showing their distribution.

Q5. Scatter Plot

Create a scatter plot showing the relationship between:

- Hours Studied
 - Exam Marks
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Q6. Sales by Region

Create a bar chart comparing sales in:

- North
 - South
 - East
 - West
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Q7. Product-wise Sales Trend

Create a line chart comparing sales of three different products over five months.

Q8. Customer Type Distribution

Create a pie chart showing the distribution of:

- New Customers
 - Returning Customers
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